

range and on-the-spot bill printing, and connect to the home area network.

### How do these help to conserve energy and reduce monthly bills?

Smart meters enable control over an individual's household energy usage and help save money. According to a report, households in the UK are already saving around Rs 2240 annually with the implementation of smart meters.

The meter comes with an LCD that displays current power consumption. It is surprising to know that even after shutting down all appliances in a household, power is still consumed through unseen sensors and various adaptors.

Smart meters help individuals keep track of their utility bills, which enables them to gauge and curb their energy consumption, leading to reduced costs and emissions at the same time.

The demand for electricity peaks at certain times during the day and during particular seasons, which leads to an increase in the cost of electricity. The meter can be tuned to identify these time-frames and adjust the power supply in the household and appliances in order to match the desired monthly electricity bill. Owing to their intelligent vigilance system, smart grids can turn on selected home appliances when power is at its cheapest mode and turn off the applications when the power is at its most expensive mode.

### Won't they be prone to hacking?

Like all other digital things, even smart meters are prone to hacking. Having said that, our effort is on to have temper-proof features in our microcontroller unit (MCU). Our MCF51EM256 device has a temper-detection feature that is connected to a real-time clock (RTC) which can be battery-backed-up even if there is no powers supply to the meter. So even if the meter is tempered, connected or not connected to the power, the RTC records the accurate time of tempering of the meter.

### What will be their cost to customers?

According to *The Economist*, a smart meter costs about \$125. This is only an indication; the final cost varies from country to country and tender to tender. Also, the installation charges can cost a few hundred dollars more depending on the mandatory communications network and data-management software. However, government subsidy is essen-

tial towards the same.

### What's your (Freescale's) contribution?

To help advance innovation in energy management, Freescale has introduced a turnkey metering reference design built upon an advanced MCU customised for the electrical metering market.

The polyphase reference design, based on Freescale's advanced MCF51EM256 ColdFire MCU, is a cost-optimised implementation of single-phase or three-phase smart electrical meter. It contains the necessary hardware and software to make development of smart meters quick and easy.

Another solution is a single-phase smart meter that features the LCD-capable 9S08LG32 microcontroller working as the metering brain, in conjunction with the MC1302 ZigBee transceiver that can be used for a host of communication activities including automated meter reading within a specified range, on-the-spot bill printing and, in the future, home area network.

### What remains to be done for smart grid to become practical?

Every country including India is facing energy problems with unreliable and outdated networks amounting to huge transmission losses each year. They are all trying to implement their own version of smart grids, and each country has a different time-frame in mind for implementation.

With different economical and political backgrounds, it is impossible to change the entire system due to enormous costs involved. Hence this technology could be implemented in a phased manner hoping that technological advancements and economies of scale would help to reduce the implementation costs.

### Finally, do the benefits of a smart grid justify the investment required in its installation?

A smart grid is a complex, comprehensive and integrated monitoring and operating system. It will provide benefits only after considerable investments have been made in upgrading the infrastructure of the nation's utilities and the monitoring and control devices in the homes and businesses of consumers. This is a long-drawn process and requires support of policy makers to effectively communicate the benefits to the customer. We should also consider the benefits that this solution will bring to the environment and society at large.

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